

Linear Time Series Assignment

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Part I: The data

Question 1: Data description

The series shows the French Industrial Production Index (IPI) for the pharmaceutical industry (NAF rev. 2, division 21). Let's note that it is already seasonally and working-day adjusted (CVS-CJO), base 100 in 2021 and published monthly by INSEE (BDM). It spans January 1990 to February 2026 (which equates to $T = 434$ observations). Division 21 covers basic pharmaceutical substances and finished medicinal products. Graphically we can see that the production follows a stable long-run upward trajectory, which is probably driven by population ageing and public health spending, interrupted by the Covid-19 shock (2020) and a subsequent vaccine-driven rebound.

Period	Jan. 1990 – Feb. 2026	Std. deviation	25.09
T	434	Min (Feb. 1990)	25.11
Mean	69.36	Max (Aug. 2024)	133.20
Median	70.92		

Table 1 Descriptive statistics : IPI Pharmaceutical industry (CVS-CJO, base 100 = 2021)

The mean (69.36) over 1990 to 2026 lies well below the 2021 reference, consistent with the long-run upward trend. Since the series is strictly positive with growing variance, we applied a logarithmic transformation $Y_t = \log X_t$. Beyond variance stabilisation, the log transform also converts multiplicative growth into additive increments, which makes it the natural scale for such a trending index.

Question 2: Stationarisation

We apply $Y_t = \log X_t$ to stabilise the variance. Since Y_t exhibits a visible upward trend, unit root tests are run with a constant and a time trend. Results are reported in Table below.

Test	Specification	Statistic	p -value
ADF	Const. + trend	-1.890	0.624
PP	Const. + trend	-31.734	< 0.01
KPSS	Trend stat. (H_0)	1.179	< 0.01
KPSS	Level stat. (H_0)	6.749	< 0.01

Table 2 Unit root tests on $Y_t = \log(IPI_t)$ in level

The ADF fails to reject the unit root hypothesis ($p = 0.624$), while the PP rejects it ($p < 0.01$). This discrepancy is well-documented: the PP test can over-reject in the presence of structural breaks, visible in our series around 2020. The KPSS rejects stationarity under both the trend and level null hypotheses ($p < 0.01$). Since ADF and KPSS agree, we conclude that $Y_t \notin I(0)$.

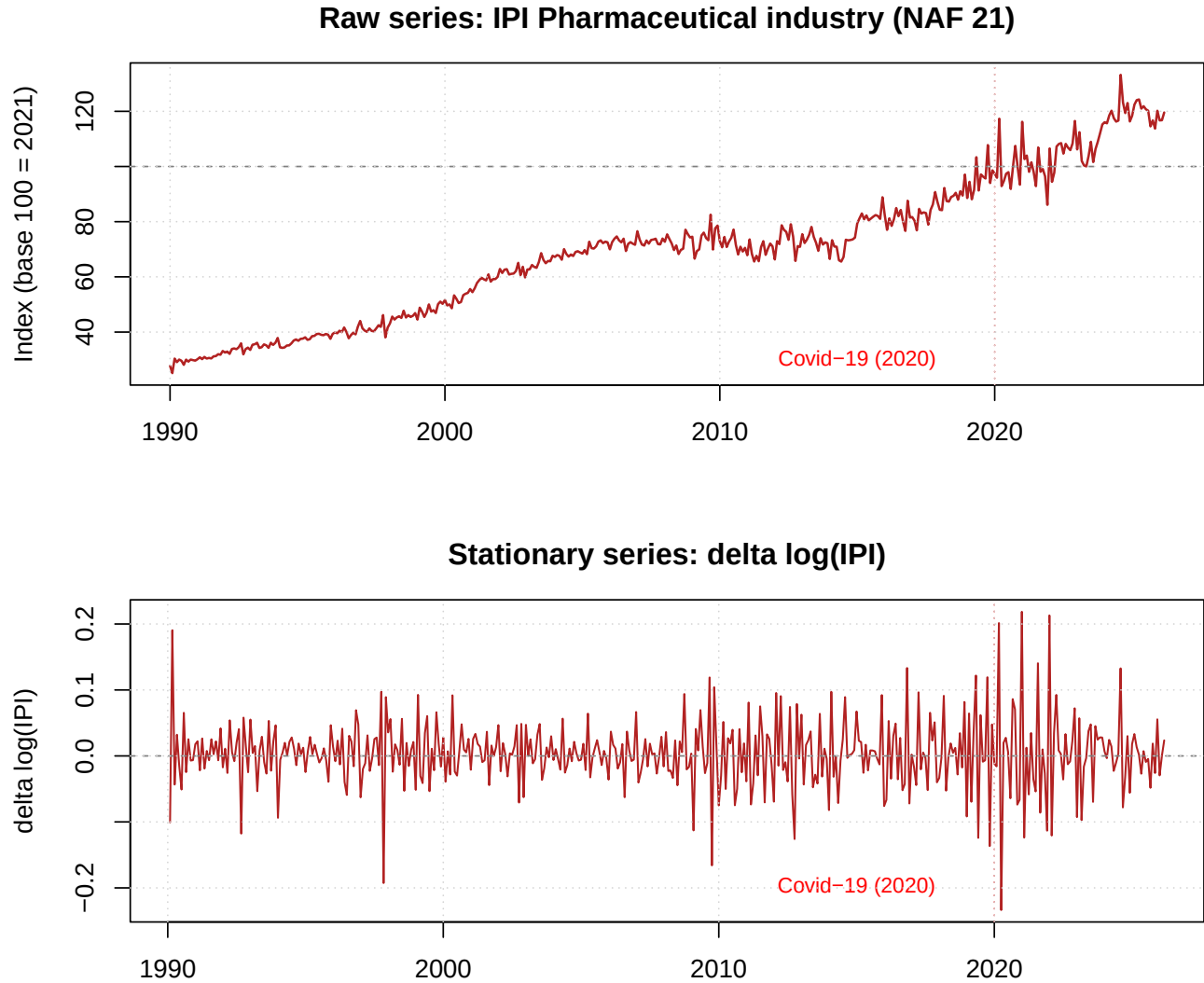
To determine the integration order, we apply a first difference $\Delta Y_t = Y_t - Y_{t-1} = \log(X_t/X_{t-1})$ and re-test. Since ΔY_t no longer exhibits a visible trend, tests are run with a constant only (as shown on table).

Test	Specification	Statistic	p -value
ADF	Constant	-9.063	< 0.01
PP	Constant	-545.6	< 0.01
KPSS	Level stat. (H_0)	0.109	> 0.10

Table 3 Unit root tests on $\Delta Y_t = \Delta \log(IPI_t)$

All three tests now agree: ADF and PP strongly reject the unit root ($p < 0.01$), and KPSS fails to reject level stationarity ($p > 0.10$), confirming $\Delta Y_t \sim I(0)$. The conjunction of $Y_t \notin I(0)$ and $\Delta Y_t \sim I(0)$ establishes $Y_t \sim I(1)$, so $d = 1$ is sufficient to render the log-transformed series stationary.

Question 3: Graphical representation



Figures 1 and 2: On top, the raw series X_t . For bottom: stationary series $\Delta Y_t = \Delta \log(IPI_t)$. The Covid-19 shock (2020) is the largest outlier in the differenced series.

Part II: ARMA models

Question 4: Identification, estimation and validation

When it comes to the identification and as shown below, the ACF cuts off after lag 1 ($\hat{\rho}(1) \approx -0.68$) while the PACF shows significant spikes at the first four lags before decaying geometrically, suggesting a MA(1) component with a possible low-order AR structure. We therefore estimate MA(1), MA(2), ARMA(1,1), ARMA(2,1), ARMA(3,1), ARMA(4,1) and ARMA(1,2) as candidates.

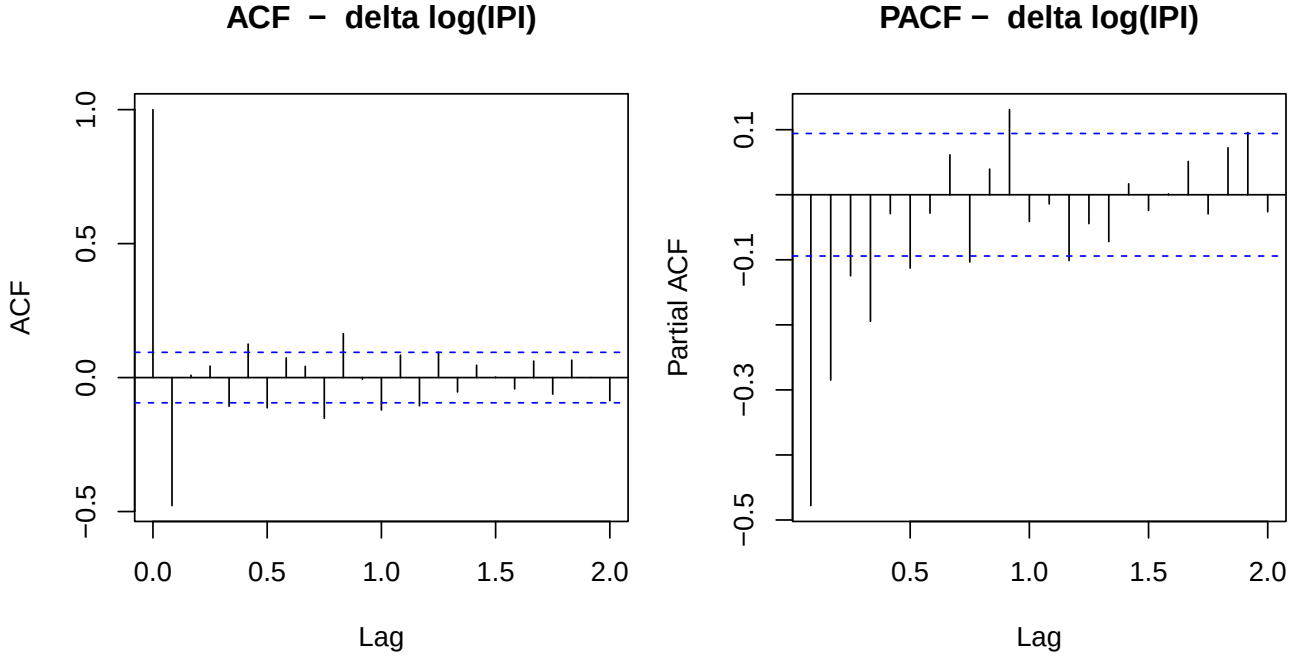


Figure 3: ACF and PACF of ΔY_t (95% bounds: $\pm 1.96/\sqrt{T}$).

Now we proceed with the selection by comparing the models that we kept to compare.

Model	AIC	BIC	Elimination rationale
MA(1)	-1503.820	-1491.608	All coefficients significant
ARMA(1,2)	-1502.518	-1482.165	All coef. sig. but BIC +9.4 vs MA(1)
ARMA(1,1)	-1501.924	-1485.642	$\hat{\phi}_1$ insig. ($t = -0.32$)
MA(2)	-1501.923	-1485.640	$\hat{\theta}_2$ insig. ($t = 0.32$)
ARMA(2,1)	-1499.933	-1479.580	$\hat{\phi}_1, \hat{\phi}_2$ insig. ($t = -0.19, 0.09$)
ARMA(4,1)	-1499.562	-1471.067	$\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3$ insig. ($ t < 1.10$)
ARMA(3,1)	-1498.144	-1473.719	$\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3$ insig. ($ t < 0.46$)
AR(2)	-1480.477	-1464.194	Dominated on both criteria
AR(1)	-1446.609	-1434.397	Dominated on both criteria

Table 4 Information criteria and elimination rationale for candidate ARMA models on ΔY_t ($T = 433$)

Both AIC and BIC unanimously select **MA(1)**. All ARMA($p, 1$) specifications with $p \geq 1$ are dominated by MA(1) on both criteria, confirming that the AR component does not contribute significantly. ARMA(1,1) and MA(2) are further eliminated by insignificant coefficients ($|t| < 0.33$). The MA(1) is therefore retained.

Thus we get the following estimation:

$$\Delta Y_t = \mu + \varepsilon_t - \theta_1 \varepsilon_{t-1}, \quad \varepsilon_t \stackrel{iid}{\sim} WN(0, \sigma^2)$$

Parameter	Estimate	Std. Error	t -stat	p -value
$\hat{\theta}_1$	-0.6841	0.0348	-19.64	< 0.001
$\hat{\mu}$	0.0033	0.0006	5.15	< 0.001
$\hat{\sigma}^2$	0.001789			

Table 5 MA(1) estimated parameters

Regarding the validity, the Ljung-Box test assesses whether the residuals are white noise (H_0 : no autocorrelation up to lag H ; H_1 : at least one autocorrelation is non-zero).

Lags H	Q'	df	p -value
5	4.526	4	0.339
10	17.520	9	0.041
15	32.145	14	0.004
20	35.092	19	0.014

Table 6 *Ljung-Box test on MA(1) residuals*

H_0 is not rejected at lag 5 ($p = 0.339$) but is rejected at longer horizons. However, this rejection reflects non-constant residual *variance* rather than residual autocorrelation in the *mean*: the ACF of squared residuals (figure) shows significant spikes, indicating ARCH effects driven by the Covid-19 outlier. Since ARIMA models only the conditional mean, modelling conditional variance is beyond the scope of this framework. The MA(1) is therefore considered valid.

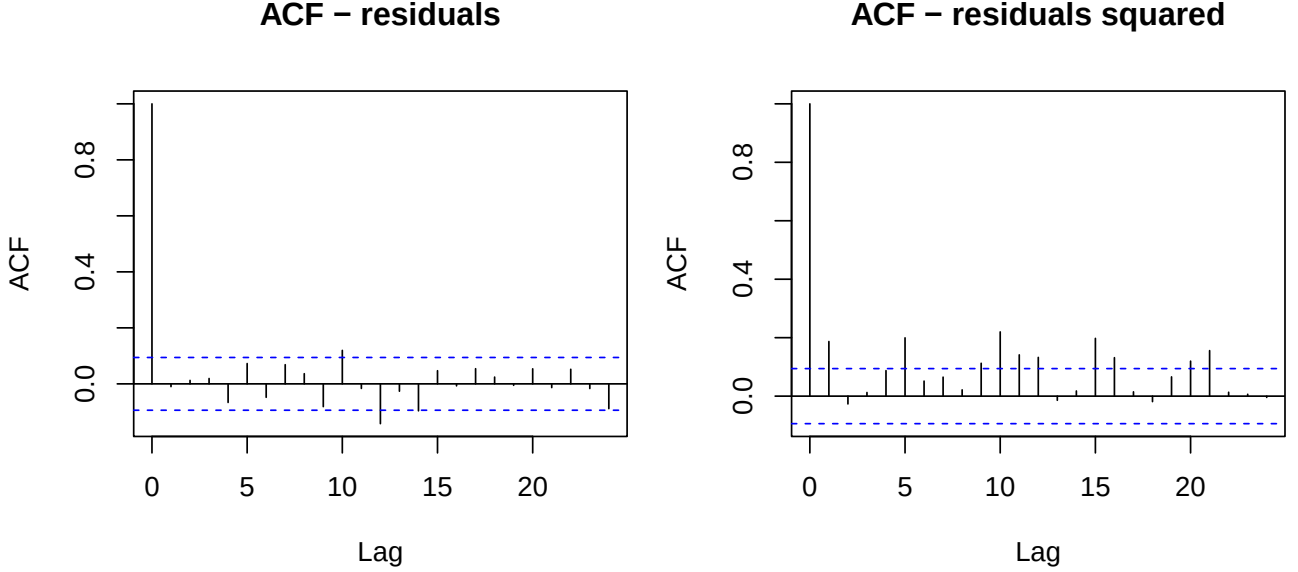


Figure 4: *Left plot: the ACF of MA(1) residuals $\hat{\varepsilon}_t$ and right one: ACF of squared residuals $\hat{\varepsilon}_t^2$*

Regarding the ACF of residuals, we can notice that most lags lie within the 95% confidence bounds, consistent with white noise in the mean. On the other hand, the ACF of squared residuals shows significant spikes which indicate conditional heteroskedasticity (ARCH effects).

Question 5: ARIMA model

Since $\Delta Y_t \sim \text{MA}(1)$ and $d = 1$, the original log-series follows **ARIMA(0,1,1)**:

$$(1 - B)Y_t = 0.0033 + (1 + 0.6841 B) \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} WN(0, 0.001789)$$

This is the EWMA model. The drift $\hat{\mu} = 0.0033$ implies an average monthly growth rate of 0.33%.

Part III: Prediction

Question 6: Confidence region on (Y_{T+1}, Y_{T+2})

The ARIMA(0,1,1) model gives h -step ahead prediction errors:

$$e_{T+1} = \varepsilon_{T+1}, \quad e_{T+2} = \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1}$$

with point forecasts $\hat{Y}_{T+h|T} = Y_T + h\hat{\mu} + \hat{\theta}_1 \hat{\varepsilon}_T$. Under Gaussianity, $(e_{T+1}, e_{T+2})^\top \sim \mathcal{N}(\mathbf{0}, \hat{\Sigma})$ with:

$$\hat{\Sigma} = \hat{\sigma}^2 \begin{pmatrix} 1 & \hat{\theta}_1 \\ \hat{\theta}_1 & 1 + \hat{\theta}_1^2 \end{pmatrix} = \begin{pmatrix} 0.001789 & 0.001224 \\ 0.001224 & 0.002626 \end{pmatrix}$$

The α -level confidence region is:

$$\left(\begin{pmatrix} Y_{T+1} \\ Y_{T+2} \end{pmatrix} - \begin{pmatrix} \hat{Y}_{T+1|T} \\ \hat{Y}_{T+2|T} \end{pmatrix} \right)^\top \hat{\Sigma}^{-1} \left(\begin{pmatrix} Y_{T+1} \\ Y_{T+2} \end{pmatrix} - \begin{pmatrix} \hat{Y}_{T+1|T} \\ \hat{Y}_{T+2|T} \end{pmatrix} \right) \leq \chi_{1-\alpha}^2(2)$$

with $\chi_{0.95}^2(2) = 5.991$ for $\alpha = 5\%$.

Question 7: Hypotheses

The hypotheses needed to get this region are the following:

1. **Gaussianity of innovations:** $\varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$, as stated in the assignment. This ensures $(e_{T+1}, e_{T+2})^\top$ is bivariate Gaussian, which is required for the χ^2 distribution of the quadratic form.
2. **Correct model specification:** the true DGP is ARIMA(0,1,1). Any misspecification of p , d or q would alter the expression of the prediction errors and invalidate Σ .
3. **Parameter estimation:** true θ_1 and σ^2 are replaced by consistent ML estimates; valid asymptotically ($T = 433$).
4. **Information set:** $\hat{Y}_{T+h|T}$ is conditioned on $\mathcal{F}_T = \sigma(Y_1, \dots, Y_T)$, and future innovations $\varepsilon_{T+1}, \varepsilon_{T+2}$ are independent of $\mathcal{F}_T$.
5. **Stationarity of ΔY_t :** confirmed in Q2, ensuring the MA(1) representation and its forecast error formula are valid.
6. **Invertibility of $\hat{\Sigma}$:** $\det(\Sigma) = \sigma^4(1 + \theta_1^2 - \theta_1^2) = \sigma^4 > 0$ always, so the quadratic form is well-defined regardless of θ_1 .

Question 8: Graphical representation ($\alpha = 95\%$)

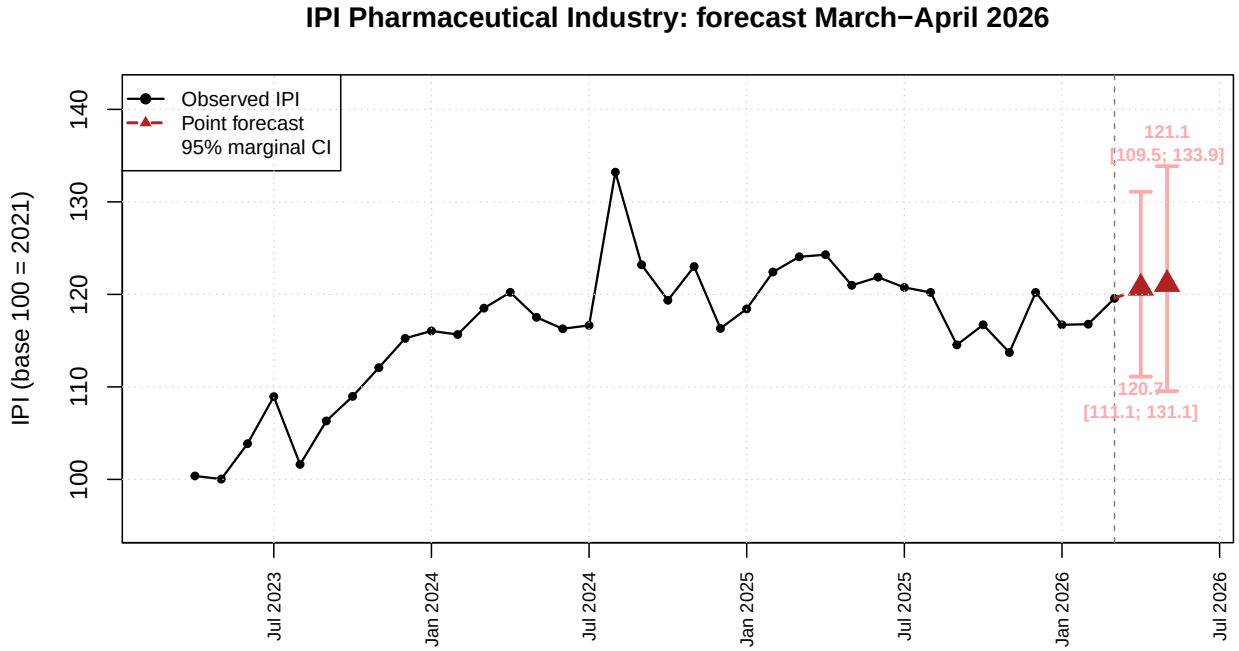


Figure 5: *ARIMA(0,1,1) point forecasts for March and April 2026 (red triangles) with marginal 95% confidence intervals (blue bars). Dashed vertical line: end of observed sample ($T = \text{Feb. 2026}$).*

Regarding the point forecasts, we get: $\hat{X}_{T+1} = 115.6$ and $\hat{X}_{T+2} = 116.0$ (base 100 = 2021), with marginal 95% CIs:

$$X_{T+1} \in [106.4; 125.6] \quad X_{T+2} \in [104.9; 128.2]$$

When it comes to the growing uncertainty : the CI for $T + 2$ is wider than for $T + 1$, reflecting $\text{Var}(e_{T+2}) = \hat{\sigma}^2(1 + \hat{\theta}_1^2) = 0.002626 > \hat{\sigma}^2 = 0.001789 = \text{Var}(e_{T+1})$. thus, forecast uncertainty grows with the horizon, as expected for any ARIMA process.

Joint region¹ : the marginal CIs displayed here are the projections of the joint elliptical confidence region derived in Q6 onto each axis. The joint region is strictly smaller than the Cartesian product of the two marginal intervals, since $\text{Cov}(e_{T+1}, e_{T+2}) = \hat{\sigma}^2 \hat{\theta}_1 = 0.001224 > 0$: a positive surprise in March 2026 tends to propagate into April 2026 through the MA(1) structure, so the two errors are not independent.

¹The chart representing the joint region has been placed in the appendix.

Level : Both forecasts are nearly identical: the MA(1) memory is exhausted after one period, so $\hat{X}_{T+2} \approx \hat{X}_{T+1}$. Both lie well below the August 2024 peak (133.2), consistent with the moderate downward correction observed since mid-2024.

Question 9: Using an Early-Available Series

Theoretical condition: Let Y_t be a stationary series with Y_{T+1} available before X_{T+1} . Knowing Y_{T+1} improves the L^2 -optimal linear forecast of X_{T+1} if and only if two conditions hold jointly, tested in a bivariate VAR(p) on $(\Delta \log X_t, Y_t)^\top$:

1. **Granger causality** (F-test): does the *past* of Y help predict X , beyond the past of X alone?
 $H_0: A_k(1, 2) = 0 \forall k$ (no Granger causality) H_1 : at least one $A_k(1, 2) \neq 0$
2. **Instantaneous causality** ($\chi^2(1)$ test): does Y_{T+1} *itself* carry information about X_{T+1} , beyond both series' histories?
 $H_0: \hat{\Sigma}_{12} = \text{Cov}(\hat{\varepsilon}_t^X, \hat{\varepsilon}_t^Y) = 0$ $H_1: \hat{\Sigma}_{12} \neq 0$

Both conditions must hold simultaneously: condition 1 ensures the *history* of Y is informative about X whereas condition 2 ensures that Y_{T+1} *itself* adds information beyond what both histories already provide. If only condition 2 holds, Y_{T+1} is useful but its past was not (which is precisely the situation we encounter empirically).

Let's now present the candidate series and hypotheses. We consider two series published before the IPI release:

Candidate 1: IPI Chimie (NAF 20), VAR(7) The chemical industry is a direct upstream supplier to pharmaceutical manufacturers. The hypothesis is that chemical production leads pharmaceutical production, so that $\Delta \log(\text{IPI Chimie})_{T+1}$ would carry predictive information about X_{T+1} . Both series are $I(1)$; we work on first-differenced logs. Lag order selected by AIC: $p = 7$.

Candidate 2: Balance of opinion on recent manufacturing production (INSEE ENQ-CONJ-ACT-IND, A10-CZ, CVS; published 2–3 weeks before the IPI). This monthly business survey asks firms about their recent production trend. Since it is published before the IPI is officially released, it satisfies the early-availability requirement by construction. The series is stationary in levels (ADF $p < 0.01$), so no differencing is required. Lag order selected by AIC: $p = 10$.

Hence, we get the following results:

Candidate Y_t	Test	Stat.	p -value	$\hat{\rho}$
IPI Chimie (NAF 20), VAR(7)	Granger (F)	1.033	0.406	−0.001
	Instantaneous (χ^2)	0.000	0.985	
Balance of opinion, VAR(10)	Granger (F)	1.793	0.058	0.103
	Instantaneous (χ^2)	4.422	0.035	

Table 7 Granger and instantaneous causality tests (H_0 not rejected if $p > 0.05$)

Regarding IPI Chimie, both H_0 's are not rejected ($p = 0.406$; $p = 0.985$), and $\hat{\rho} \approx 0$. The hypothesis that chemical production leads pharmaceutical production is not supported: pharmaceutical output is demand-driven (prescriptions, health policy) rather than input-constrained.

For Balance of opinion, Granger causality is not rejected ($p = 0.058$), but instantaneous causality is ($p = 0.035$, $\hat{\rho} = 0.103$): condition 2 holds. Knowing the survey outcome for month $T+1$ allows a marginal forecast improvement (variance reduced by $1 - 0.103^2 = 98.9\%$). Both series capture the same monthly production cycle from different measurement angles, generating contemporaneous co-movement without strong Granger-causal dynamics.

Appendix

Joint region of question 8

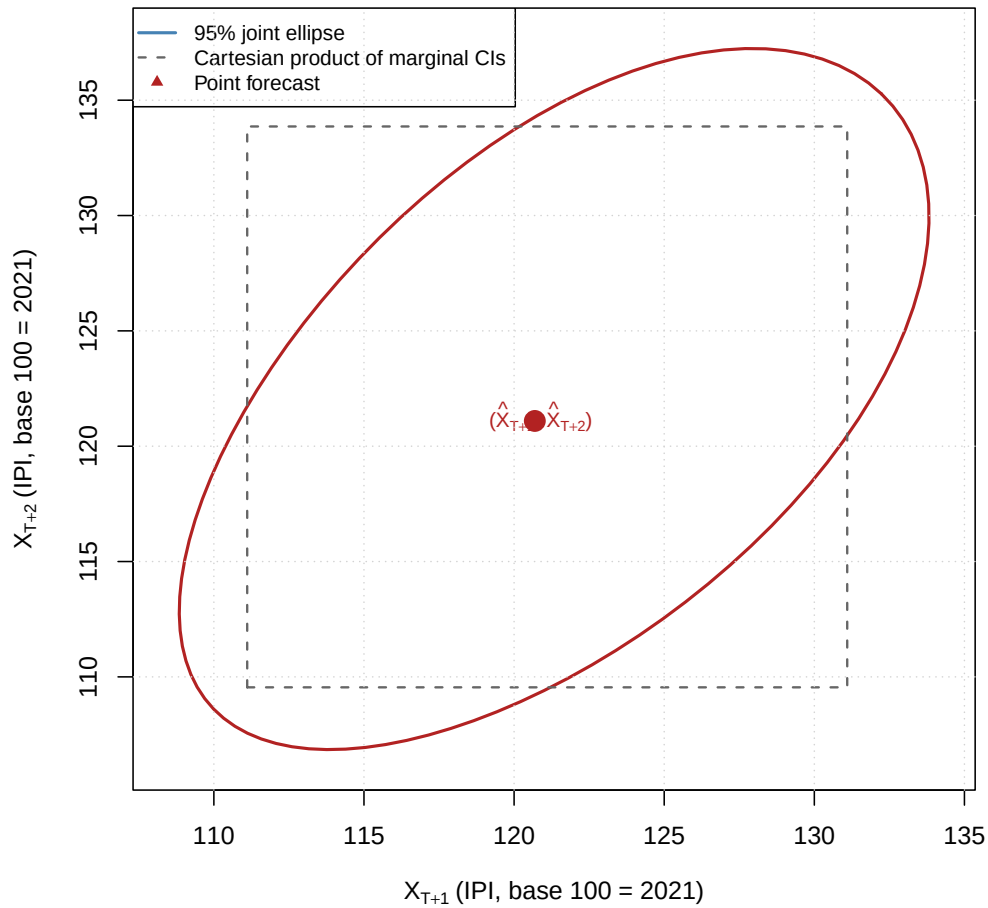


Figure 6: Joint confidence region for (X_{T+1}, X_{T+2})

R script

```
# DM Linear Time Series
# IPI Pharmaceutical Industry (NAF 21)
# done by Mathieu Tassel & Ralph Nader

# loading all required Packages
pkgs <- c("insee", "dplyr", "stringr", "tseries", "forecast", "vars", "ellipse")
for (p in pkgs) {
  if (!requireNamespace(p, quietly = TRUE)) install.packages(p)
}

library(insee); library(dplyr); library(stringr)
library(tseries); library(forecast); library(vars)

filter <- dplyr::filter
select <- dplyr::select

# Question 1) Data retrieval and descriptive statistics

idbank_list <- get_idbank_list("IPI-2021") |>
  filter(FREQ == "M") |>
  filter(NATURE == "INDICE") |>
  filter(CORRECTION == "CVS-CJO") |>
  filter(str_detect(NAF2, "~21$")) |>
  add_insee_title()

idbank_pharma <- idbank_list |> pull(idbank)

raw <- get_insee_idbank(idbank_pharma)
```

```

ipi_pharma <- raw |>
  transmute(date = as.Date(paste0(TIME_PERIOD, "-01")),
            ipi = as.numeric(OBS_VALUE)) |>
  arrange(date) |>
  filter(!is.na(ipi))

cat("\nQuestion 1) Descriptive statistics\n")
cat("Period:", format(min(ipi_pharma$date), "%Y-%m"),
    "to", format(max(ipi_pharma$date), "%Y-%m"), "\n")
cat("T:", nrow(ipi_pharma), "\n")
cat("Mean:", round(mean(ipi_pharma$ipi), 2), "\n")
cat("Median:", round(median(ipi_pharma$ipi), 2), "\n")
cat("StdDev:", round(sd(ipi_pharma$ipi), 2), "\n")
cat("Min:", round(min(ipi_pharma$ipi), 2),
    "(", format(ipi_pharma$date[which.min(ipi_pharma$ipi)], "%Y-%m"), ") \n")
cat("Max:", round(max(ipi_pharma$ipi), 2),
    "(", format(ipi_pharma$date[which.max(ipi_pharma$ipi)], "%Y-%m"), ") \n")
cat("CV:", round(sd(ipi_pharma$ipi) / mean(ipi_pharma$ipi) * 100, 1), "%\n")

write.csv(ipi_pharma, "ipi_pharma_cvs_cjo.csv", row.names = FALSE)

# Question 2) Stationarisation

ipi_ts <- ts(ipi_pharma$ipi, start = c(1990, 1), frequency = 12)
log_ipi <- log(ipi_ts)
d_log_ipi <- diff(log_ipi)

cat("\nQuestion 2) Unit root tests on log(IPI) in level\n")
print(adf.test(log_ipi, alternative = "stationary"))
print(pp.test(log_ipi, alternative = "stationary"))
print(kpss.test(log_ipi, null = "Trend"))
print(kpss.test(log_ipi, null = "Level"))

cat("\nQuestion 2) Unit root tests on diff(log(IPI))\n")
print(adf.test(d_log_ipi, alternative = "stationary"))
print(pp.test(d_log_ipi, alternative = "stationary"))
print(kpss.test(d_log_ipi, null = "Level"))

# Question 3) Graphical representation

par(mfrow = c(2, 1), mar = c(4, 4, 3, 1))

plot(ipi_ts,
     main = "Raw series: IPI Pharmaceutical industry (NAF 21)",
     ylab = "Index (base 100 = 2021)", xlab = "",
     col = "#B22222", lwd = 1.5)
abline(h = 100, lty = 2, col = "grey50")
abline(v = 2020, lty = 3, col = "red")
text(2015, 30, "Covid-19 (2020)", col = "red", cex = 0.8)
grid()

plot(d_log_ipi,
     main = "Stationary series: delta log(IPI)",
     ylab = "delta log(IPI)", xlab = "",
     col = "#B22222", lwd = 1.2)
abline(h = 0, lty = 2, col = "grey50")
abline(v = 2020, lty = 3, col = "red")
text(2015, min(d_log_ipi) * 0.85, "Covid-19 (2020)",
     col = "red", cex = 0.8)
grid()

dev.copy(pdf, "figure_q3.pdf", width = 8, height = 7); dev.off()

# Question 4) Identification

# ACF / PACF
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
acf(as.numeric(d_log_ipi), lag.max = 24, main = "ACF of delta log(IPI)")
pacf(as.numeric(d_log_ipi), lag.max = 24, main = "PACF of delta log(IPI)")
dev.copy(pdf, "figure_acf_pacf.pdf", width = 8, height = 4); dev.off()

# Candidate models (from ACF/PACF reading)
res_01 <- arima(d_log_ipi, order = c(0, 0, 1)) # MA(1) i.e. the baseline
res_02 <- arima(d_log_ipi, order = c(0, 0, 2)) # MA(2)
res_11 <- arima(d_log_ipi, order = c(1, 0, 1)) # ARMA(1,1)
res_21 <- arima(d_log_ipi, order = c(2, 0, 1)) # ARMA(2,1)
res_31 <- arima(d_log_ipi, order = c(3, 0, 1)) # ARMA(3,1)
res_41 <- arima(d_log_ipi, order = c(4, 0, 1)) # ARMA(4,1)
res_12 <- arima(d_log_ipi, order = c(1, 0, 2)) # ARMA(1,2)
res_20 <- arima(d_log_ipi, order = c(2, 0, 0)) # AR(2)
res_10 <- arima(d_log_ipi, order = c(1, 0, 0)) # AR(1)

# AIC / BIC table
models <- list("MA(1)" = res_01, "ARMA(1,2)" = res_12,
              "ARMA(1,1)" = res_11, "MA(2)" = res_02,
              "ARMA(2,1)" = res_21, "ARMA(4,1)" = res_41,
              "ARMA(3,1)" = res_31, "AR(2)" = res_20,
              "AR(1)" = res_10)

ic_table <- data.frame(
  Model = names(models),

```



```

AIC = round(sapply(models, AIC), 3),
BIC = round(sapply(models, BIC), 3)
)
cat("\nQuestion 4) AIC/BIC (sorted by AIC)\n")
print(ic_table[order(ic_table$AIC), ])

# Coefficient significance check
check_coef <- function(nom, mod) {
  coefs <- mod$coef
  ses <- sqrt(diag(mod$var.coef))
  t_stats <- coefs / ses
  p_vals <- 2 * (1 - pnorm(abs(t_stats)))
  cat("\n====", nom, "====\n")
  print(round(data.frame(Estimate = coefs, SE = ses,
                        t_stat = t_stats, p_value = p_vals), 4))
  cat("AIC:", round(AIC(mod), 3), "| BIC:", round(BIC(mod), 3), "\n")
}

check_coef("MA(1)", res_01)
check_coef("ARMA(1,2)", res_12)
check_coef("ARMA(1,1)", res_11)
check_coef("MA(2)", res_02)
check_coef("ARMA(2,1)", res_21)
check_coef("ARMA(4,1)", res_41)
check_coef("ARMA(3,1)", res_31)

# Retained model
best_model <- res_01
cat("\n>>> RETAINED MODEL: MA(1)\n")

# Ljung-Box on MA(1) residuals
cat("\nQuestion 4) Ljung-Box test on MA(1) residuals\n")
for (h in c(5, 10, 15, 20)) {
  lb <- Box.test(residuals(best_model), lag = h,
                 type = "Ljung-Box", fitdf = 1)
  sig <- ifelse(lb$p.value < 0.05, "\n*", "")
  cat("lag=", formatC(h, width = 2), "| Q'=",
      round(lb$statistic, 3), "| df=", lb$parameter,
      "| p=", round(lb$p.value, 4), sig, "\n")
}

# Residual diagnostics figure (ACF + ACF squared)
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
acf(as.numeric(residuals(best_model)),
    lag.max = 24, main = "ACF - residuals")
acf(as.numeric(residuals(best_model))^2,
    lag.max = 24, main = "ACF - residuals squared")
dev.copy(pdf, "figure_residuals_MA1.pdf", width = 8, height = 3.5)
dev.off()

# Final parameter recap
coefs <- best_model$coef
ses <- sqrt(diag(best_model$var.coef))
t_stats <- coefs / ses

cat("\nQuestion 4) MA(1) parameter recap\n")
cat("theta1:", round(coefs["ma1"], 4),
    "| SE:", round(ses["ma1"], 4),
    "| t:", round(t_stats["ma1"], 4), "\n")
cat("mu:", round(coefs["intercept"], 4),
    "| SE:", round(ses["intercept"], 4),
    "| t:", round(t_stats["intercept"], 4), "\n")
cat("sigma^2:", round(best_model$sigma2, 6), "\n")
cat("AIC:", round(AIC(best_model), 3), "\n")
cat("BIC:", round(BIC(best_model), 3), "\n")

# Q8) Forecast and joint confidence region

theta1 <- as.numeric(-best_model$coef["ma1"])
mu <- as.numeric(best_model$coef["intercept"])
sig2 <- as.numeric(best_model$sigma2)
eps_T <- as.numeric(tail(residuals(best_model), 1))
Y_T <- as.numeric(tail(log_ipi, 1))

Yhat_T1 <- Y_T + mu + theta1 * eps_T
Yhat_T2 <- Y_T + 2 * mu + theta1 * eps_T

cat("\n==== Q8 Point forecasts ==== \n")
cat("Yhat_{T+1|T}:", round(Yhat_T1, 4), "-> IPI:", round(exp(Yhat_T1), 2), "\n")
cat("Yhat_{T+2|T}:", round(Yhat_T2, 4), "-> IPI:", round(exp(Yhat_T2), 2), "\n")

# Covariance matrix
Sigma <- sig2 * matrix(c(1, theta1, theta1, 1 + theta1^2), nrow = 2, byrow = TRUE)
cat("Sigma:\n"); print(round(Sigma, 6))

# Marginal 95% CIs (log scale then exponentiated)
ic1_lo <- Yhat_T1 - qnorm(0.975) * sqrt(sig2)
ic1_hi <- Yhat_T1 + qnorm(0.975) * sqrt(sig2)
ic2_lo <- Yhat_T2 - qnorm(0.975) * sqrt(sig2 * (1 + theta1^2))
ic2_hi <- Yhat_T2 + qnorm(0.975) * sqrt(sig2 * (1 + theta1^2))

```

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cat("95%CIX{T+1}:[" , round(exp(ic1_lo), 2), ";", round(exp(ic1_hi), 2), "]\n")
cat("95%CIX{T+2}:[" , round(exp(ic2_lo), 2), ";", round(exp(ic2_hi), 2), "]\n")

Xhat_T1 <- exp(Yhat_T1); Xhat_T2 <- exp(Yhat_T2)
Xic1_lo <- exp(ic1_lo); Xic1_hi <- exp(ic1_hi)
Xic2_lo <- exp(ic2_lo); Xic2_hi <- exp(ic2_hi)

# ---- Figure 1: time-series forecast ----
n_show <- 36
ipi_show <- tail(ipi_ts, n_show)
dates_num <- as.numeric(time(tail(ipi_ts, n_show)))
date_T1 <- max(time(ipi_ts)) + 1/12
date_T2 <- max(time(ipi_ts)) + 2/12

ylim_range <- range(c(ipi_show, Xic1_lo, Xic1_hi, Xic2_lo, Xic2_hi))

pdf("figure_forecast_q8.pdf", width = 9, height = 5)
par(mfrow = c(1, 1), mar = c(6, 5, 4, 2))

plot(dates_num, as.numeric(ipi_show),
     type = "o", pch = 19, cex = 0.6, lwd = 1.5, col = "black",
     xlim = c(min(dates_num) - 0.1, date_T2 + 0.12),
     ylim = c(ylim_range[1] - 15, ylim_range[2] + 10),
     xlab = "", ylab = "IPI(base100=2021)",
     main = "IPIPharmaceutical Industry: forecastMarch-April 2026",
     xaxt = "n")

axis(1, at = seq(2023, 2026.5, by = 0.5),
     labels = c("Jan2023", "Jul2023", "Jan2024", "Jul2024",
                "Jan2025", "Jul2025", "Jan2026", "Jul2026"),
     cex.axis = 0.75, las = 2)

abline(v = max(dates_num), lty = 2, col = "grey50")

arrows(date_T1, Xic1_lo, date_T1, Xic1_hi, angle = 90, code = 3,
       length = 0.07, col = "#FFAAAA", lwd = 2.5)
arrows(date_T2, Xic2_lo, date_T2, Xic2_hi, angle = 90, code = 3,
       length = 0.07, col = "#FFAAAA", lwd = 2.5)

lines(c(max(dates_num), date_T1, date_T2),
      c(as.numeric(tail(ipi_ts, 1)), Xhat_T1, Xhat_T2),
      col = "#B22222", lwd = 2, lty = 2)
points(c(date_T1, date_T2), c(Xhat_T1, Xhat_T2),
       pch = 17, col = "#B22222", cex = 1.8)

# T+1 label below, T+2 label above
text(date_T1, Xic1_lo - 5,
     paste0(round(Xhat_T1, 1), "\n[" , round(Xic1_lo, 1), ";", round(Xic1_hi, 1), "]",
     col = "#FFAAAA", cex = 0.78, font = 2, adj = 0.5)
text(date_T2, Xic2_lo + 4,
     paste0(round(Xhat_T2, 1), "\n[" , round(Xic2_lo, 1), ";", round(Xic2_hi, 1), "]",
     col = "#FFAAAA", cex = 0.78, font = 2, adj = 0.5)

legend("topleft",
      legend = c("ObservedIPI", "Pointforecast", "95%marginalCI"),
      col = c("black", "#B22222", "#FFAAAA"),
      lty = c(1, 2, NA), pch = c(19, 17, NA),
      lwd = c(1.5, 2, 2.5), cex = 0.82, bg = "white")
grid()
dev.off()

ell <- ellipse(Sigma, centre = c(Yhat_T1, Yhat_T2), level = 0.95)
ell_ipi <- exp(ell)

# Cartesian product of marginal CIs (rectangle)
rect_x <- c(Xic1_lo, Xic1_hi, Xic1_hi, Xic1_lo, Xic1_lo)
rect_y <- c(Xic2_lo, Xic2_lo, Xic2_hi, Xic2_hi, Xic2_lo)

pdf("figure_ellipse_q8.pdf", width = 7, height = 7)
par(mfrow = c(1, 1), mar = c(5, 5, 4, 2))

plot(ell_ipi[, 1], ell_ipi[, 2],
     type = "l", lwd = 2, col = "#B22222",
     xlim = range(c(ell_ipi[, 1], rect_x)) + c(-0.5, 0.5),
     ylim = range(c(ell_ipi[, 2], rect_y)) + c(-0.5, 0.5),
     xlab = expression(X[T+1] ~ "(IPI(base100=2021))"),
     ylab = expression(X[T+2] ~ "(IPI(base100=2021))"),

# Marginal CI rectangle
lines(rect_x, rect_y, lty = 2, col = "grey40", lwd = 1.5)

# Point forecast
points(Xhat_T1, Xhat_T2, pch = 21, bg = "#B22222", col = "#B22222", cex = 1.8)
text(Xhat_T1 + 0.2, Xhat_T2 + 0.2,
     expression(paste("(", hat(X)[T+1], ", ", hat(X)[T+2], ")")),
     col = "#B22222", cex = 0.85, font = 2)

legend("topleft",
      legend = c("95%jointellipse", "CartesianproductofmarginalCIs", "Pointforecast"),
      col = c("steelblue", "grey40", "#B22222"),
      lty = c(1, 2, NA), pch = c(NA, NA, 17),
      lwd = c(2, 1.5, NA), cex = 0.82, bg = "white")

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grid()
dev.off()

# Question 9) Granger causality tests

# Candidate nb 1: IPI Chimie (NAF 20)
raw_ch <- get_insee_idbank("010767783")

ipi_chimie <- raw_ch |>
  transmute(date = as.Date(paste0(TIME_PERIOD, "-01")),
            ipi = as.numeric(OBS_VALUE)) |>
  arrange(date) |>
  filter(!is.na(ipi))

write.csv(ipi_chimie, "ipi_chimie_naf20.csv", row.names = FALSE)

merged <- inner_join(ipi_pharma |> rename(pharma = ipi),
                    ipi_chimie |> rename(chimie = ipi),
                    by = "date")

d_log_pharma <- diff(log(merged$pharma))
d_log_chimie <- diff(log(merged$chimie))
bivar <- cbind(d_log_pharma, d_log_chimie)
colnames(bivar) <- c("pharma", "chimie")

p1 <- as.integer(VARselect(bivar, lag.max = 12,
                          type = "const")$selection["AIC(n)"])
var1 <- VAR(bivar, p = p1, type = "const")
g1 <- causality(var1, cause = "chimie")
s1 <- cov(resid(var1))
rho1 <- s1[1,2] / sqrt(s1[1,1] * s1[2,2])

cat("\nQuestion 9) Candidate nb 1: IPI Chimie, VAR(", p1, ") \n")
print(g1$Granger); print(g1$Instant)
cat("Rho=", round(rho1, 4), "\n")

# Candidate nb 2: balance of opinion (ENQ-CONJ-ACT-IND)
raw_enq <- get_insee_idbank("001586064")

enq_prod <- raw_enq |>
  transmute(date = as.Date(paste0(TIME_PERIOD, "-01")),
            solde = as.numeric(OBS_VALUE)) |>
  arrange(date) |>
  filter(!is.na(solde))

write.csv(enq_prod, "balance_opinion_enq_conj.csv", row.names = FALSE)

merged2 <- inner_join(ipi_pharma |> rename(pharma = ipi),
                    enq_prod, by = "date")

d_log_pharma2 <- diff(log(merged2$pharma))
solde_aligned <- merged2$solde[-1]
bivar2 <- cbind(d_log_pharma2, solde_aligned)
colnames(bivar2) <- c("pharma", "solde")

p2 <- as.integer(VARselect(bivar2, lag.max = 12,
                          type = "const")$selection["AIC(n)"])
var2 <- VAR(bivar2, p = p2, type = "const")
g2 <- causality(var2, cause = "solde")
s2 <- cov(resid(var2))
rho2 <- s2[1,2] / sqrt(s2[1,1] * s2[2,2])

cat("\nQuestion 9) Candidate nb 2: Balance of opinion, VAR(", p2, ") \n")
print(g2$Granger); print(g2$Instant)
cat("Rho=", round(rho2, 4), "\n")
cat("Variance reduction (1 - rho^2)=", round(1 - rho2^2, 4), "\n")

```